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CS2024 - Theory of Computation

The beginning

Angel Napa

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1.

Presentación
Curso y Docente

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The *professor*

- **Name:** Angel Napa
- **Degree:** Master in Mathematics: Institute of Mathematics Pure and Applied(IMPA), Brasil
- **Area:** Combinatorics, Algorithms
- **Group:** ALCOMB
- **Subarea:**
  - Combinatorics Optimization
  - Analysis of Algorithms
  - Graph Theory
  - Operations Research

Evaluation *System*

Evaluación	Teoría	Laboratorio
	3 exámenes **(E)**: Examen **E1** (25%) Examen **E2** (25%) Examen **E3** (25%)	Evaluación continua **C1** (25%)
	75%	25%
	100%	

¿What is *Theory of Computation?*

- What are the **fundamental** capabilities and limitations of computers?
  - 1930: What is computation? - Mathematical Logicians

Three areas *interpretations*

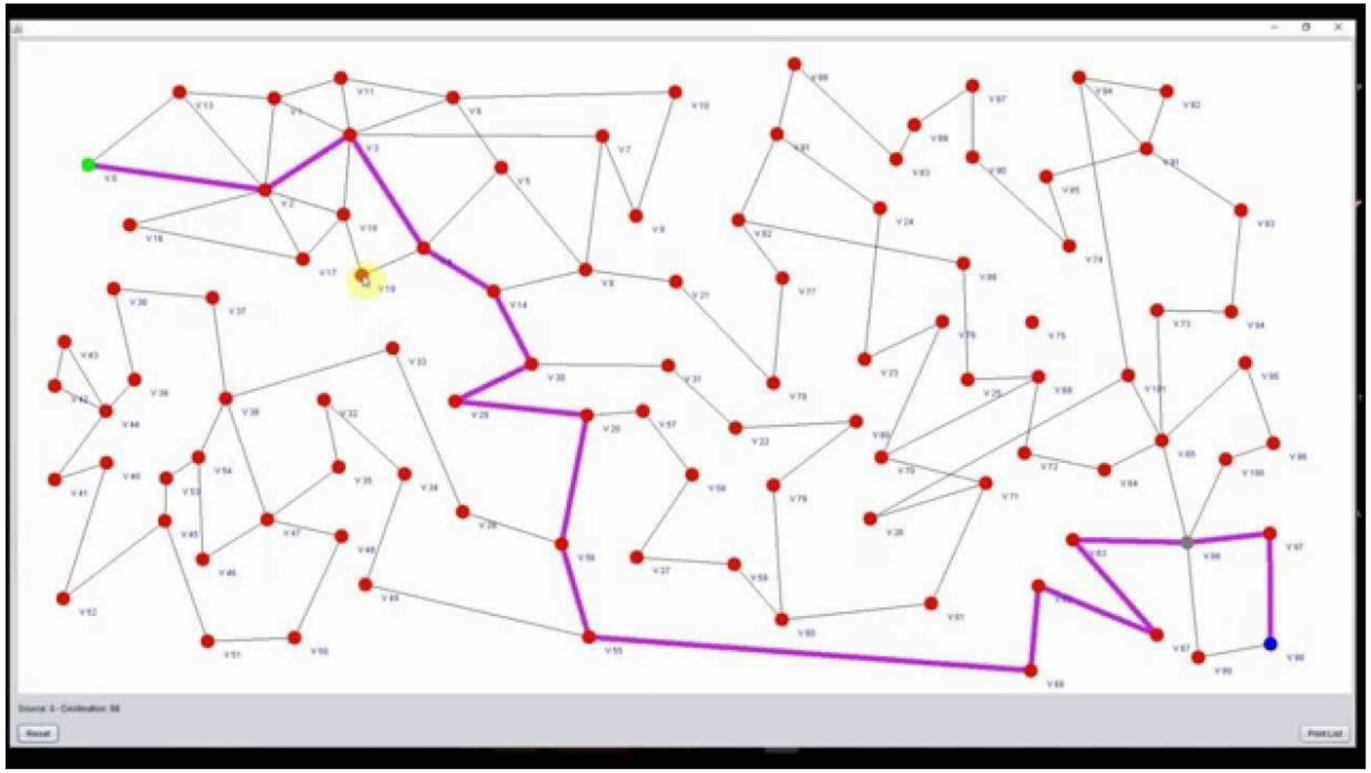
- **Complexity Theory**
  - What makes a problem computationally hard or easy?
- **Computability Theory**
  - What makes a problem computationally solvable?
- **Automata Theory**
  - First idea of an abstract simple machine

Complexity *Theory*

- **Hard vs easy**
  - Easy problem: sorting elements
  - Hard problem: Create a Valid Schedule for a University
  - What makes a problem computationally solvable?
- **WHAT** makes some problems easier and others hard?

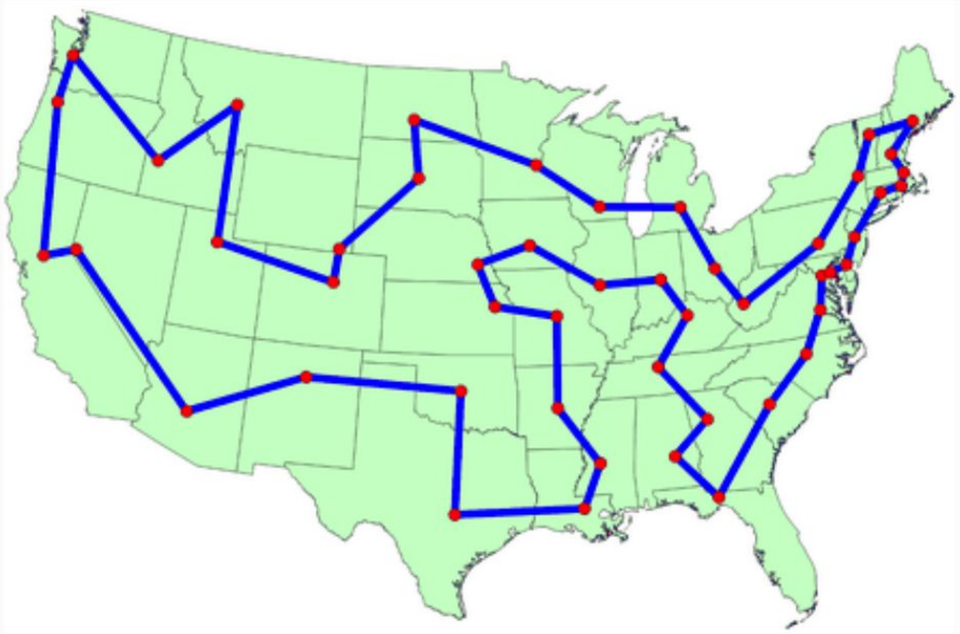
Complexity *Theory*

- Some **easy** problems
- (almost) **All AED problems**
  - Sorting
  - Shortest Path
  - Minimum Span Tree
  - etc.



Complexity *Theory*

- Some **hard** problems
  - TSP
  - Hamilton Cycle
  - Scheduling
  - 3SAT



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Complexity *Theory*

- Some **hard** problems
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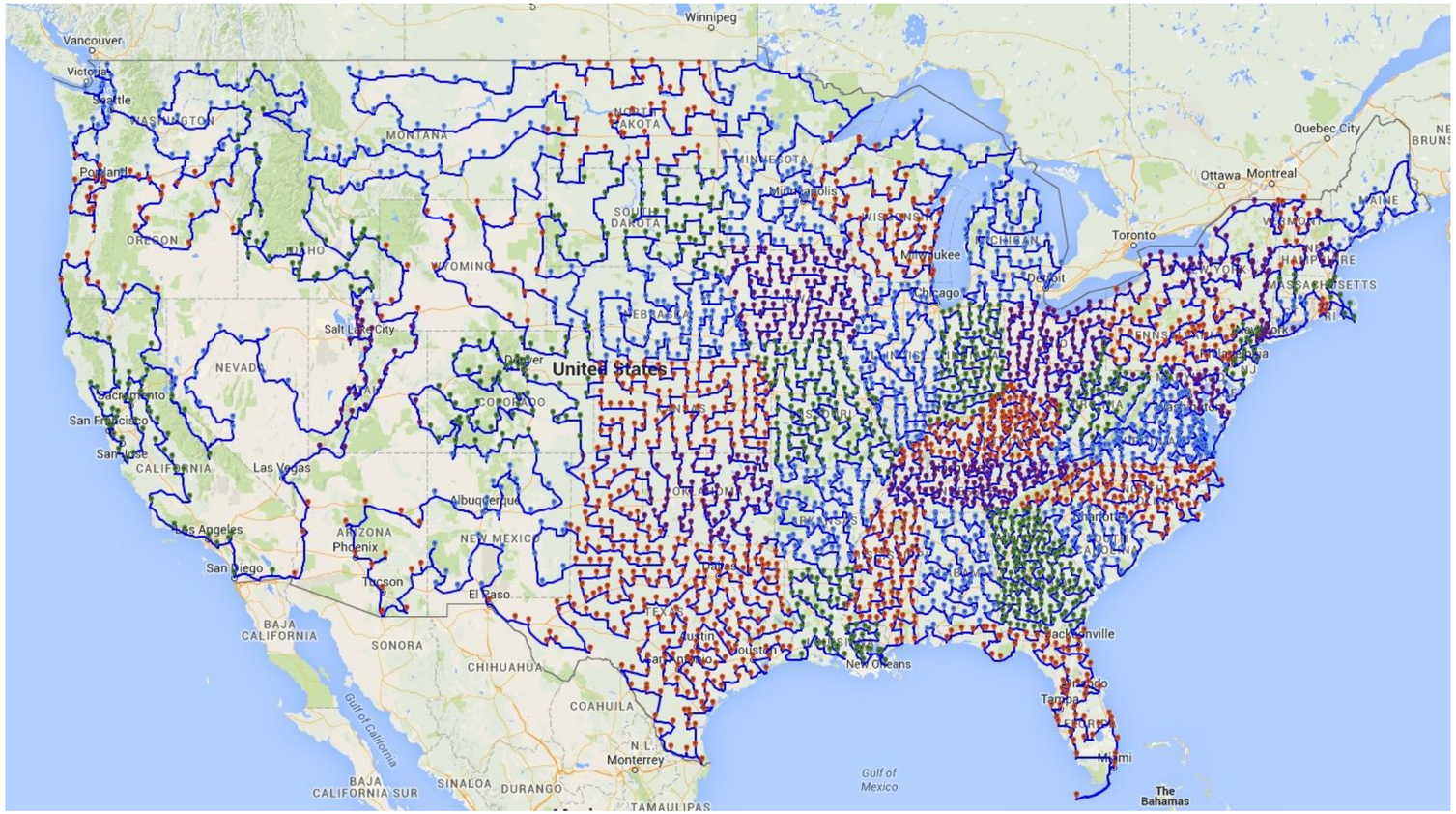
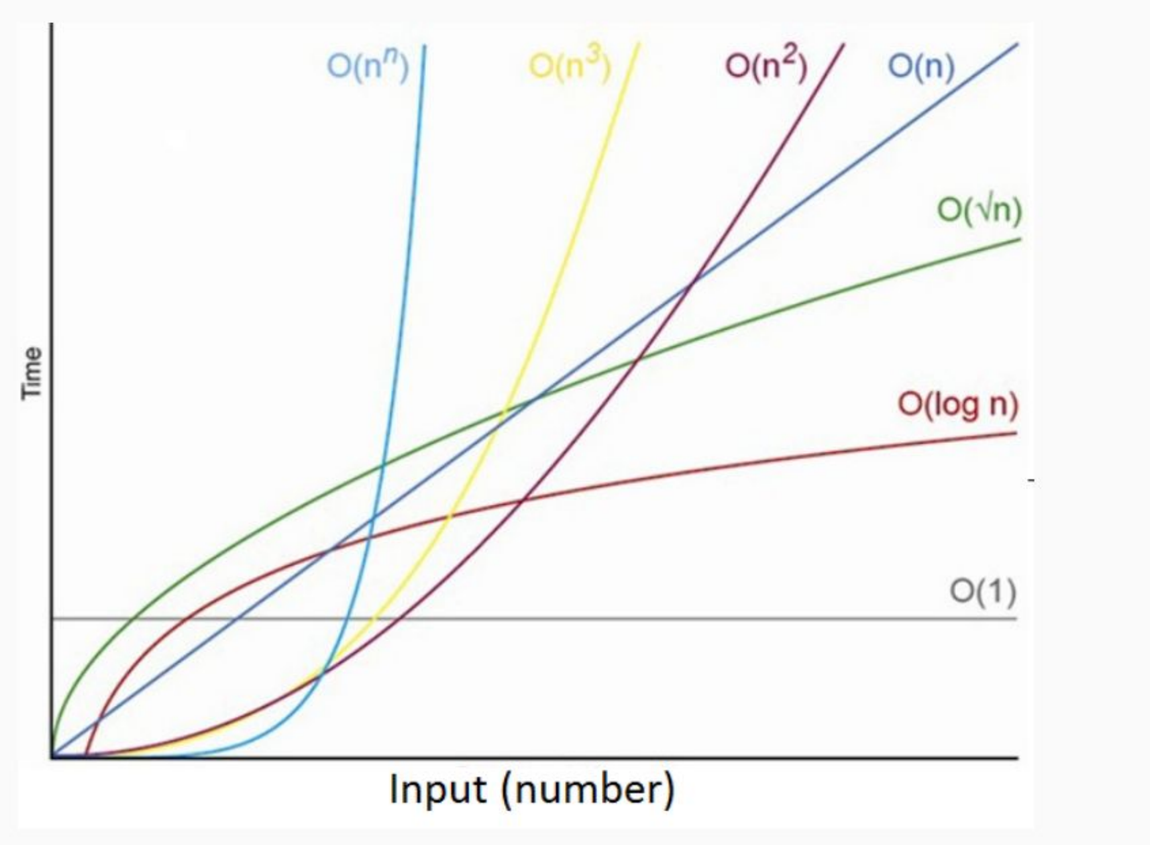
  - 3SAT



P *vs NP*



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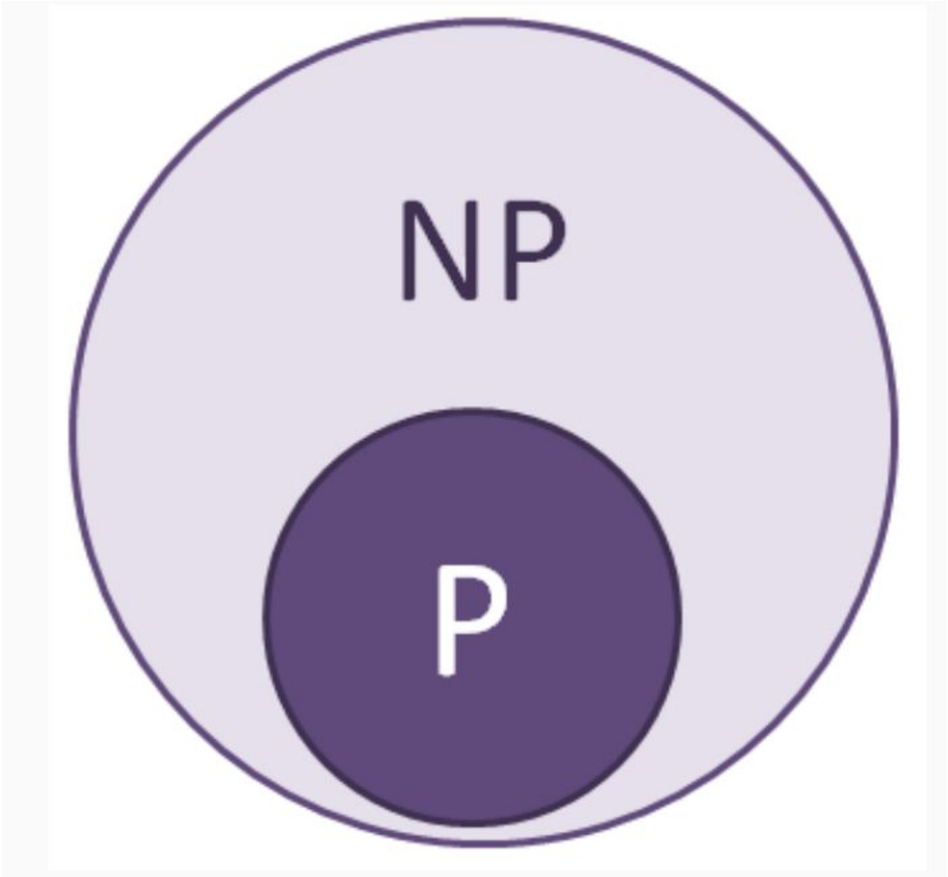
P ⊆ NP, *trivial*



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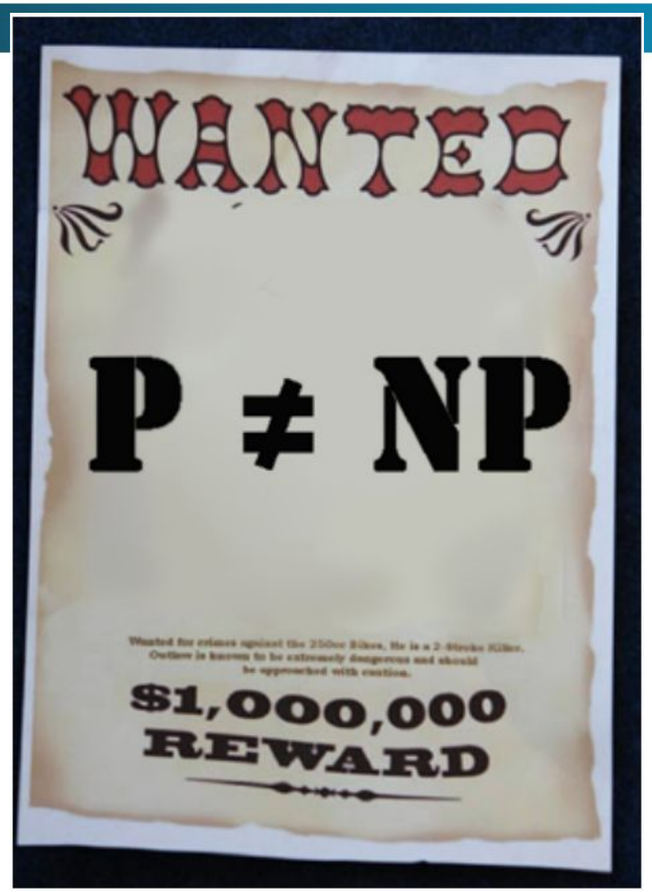
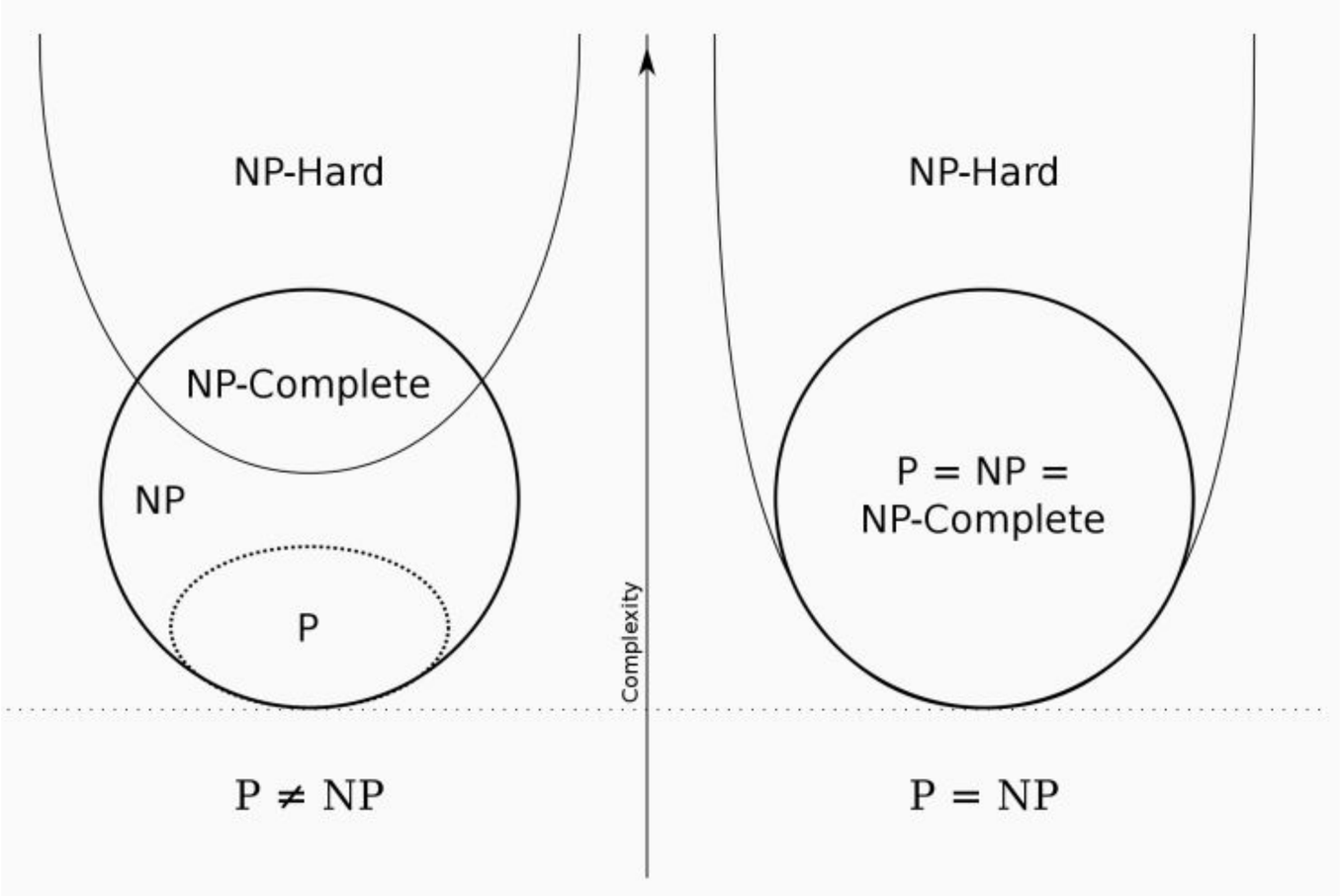
P = NP?



P = NP?


NP-Hard
NP-Complete
NP
P
Complexity
NP-Hard
P = NP =
NP-Complete
P ≠ NP
P = NP

Computability *Theory*

- **Solvable vs unsolvable**
  - Solvable problems: All AED (and more)
  - Unsolvable problem:
    - Determine the veracity of a mathematical expression
    - Determine if a program stops after receive any input
- **The Halting problem**

Automata *Theory*

- **Study models of computation**
  - Finite Automaton
  - Pushdown Automaton
  - Context Free Grammar
  - Turing Machines
- They are all different models...
  - ***or are they?***

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Quick review

You should already know this

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Mathematical Notations

- **Set**
  - **components:** elements
  - subset, proper subset, power set
  - **Types:**
    - multiset, infinite set, empty set, singleton set, unordered pair.
  - **Operations:**
    - Union, intersection, complement

Mathematical Notations

- **Sequence**
  - **components:** elements
  - subset, proper subset, power set
  - **Types:**
    - Finite(tuples):
      - k-tuples
      - ordered pair
    - Infinite

- **Sequence & Sets**
  - Cartesian product

Mathematical Notations

- **Function**(mapping)
  - Domain, Range
  - **Operations:**
    - Sum, rest, multiplication
    - Cartesian product

Mathematical Notations

- **Graph**
  - Undirected vs Directed
  - Node, edges
  - degree
  - subgraph

Mathematical Notations

- **Strings**
  - **Alphabet(Σ):** Any nonempty finite set
  - **Symbols:** elements of the alphabet
  - **String over an Σ (ω):** Finite sequence of symbols from Σ
    - **Length (|ω|):**
    - **Empty string: ε**
    - **Operations**
      - Reverse string ω^R
      - Concatenation
    - **Substring, Prefix,**

Mathematical Notations

- **Strings**
  - **Σ^1:** Strings with size 1
  - **Σ^n:** Strings with size n
  - **Σ^+ :** $\Sigma^1 \cup \Sigma^2 \cup \Sigma^3 \cup \ldots$
  - **Σ^* :** $\Sigma^0 \cup \Sigma^+$
  - **Language (L):**
    - A set of strings
    - Subset of Σ^*

Mathematical Proofs

- **Proofs**
  - **Direct Proof:**
    - P→Q
  - **Proof by induction:**
    - P(1), then P(k) → P(k+1) for all k
  - **Proof by contradiction:**
    - Assume P is false, then reach a contradiction
  - **Pigeonhole Principle:**
    - +n pigeons in just n pigeonholes? At least 2 share the same pigeon

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Thanks

Let’s keep finding mistakes

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